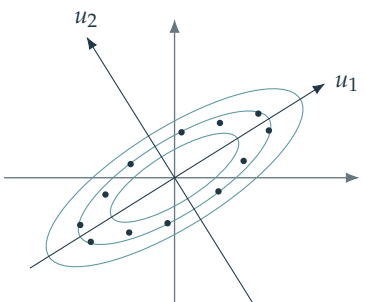
Data covariance

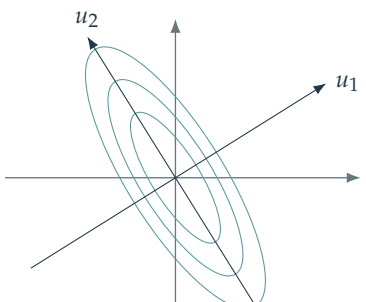
Width along $u_i \propto \sqrt{\lambda_i}$



$C/n = U \operatorname{diag}(\lambda_i) U^\top$

Quadratic level sets

Width along $u_i \propto 1/\sqrt{\lambda_i}$



$\langle Cx, x \rangle / n = \text{constant}$

Semiaxes: $r\sqrt{\lambda_i}$ (left), $r\sqrt{\lambda_1 \lambda_2}/\sqrt{\lambda_i}$ (right), $r = 1, \frac{3}{2}, 2$.